TITLE PAGE

- **Problem Statement ID –SIH26180**
- **Problem Statement Title-**A field-deployable AI-powered Smart Farming Assistant
- **Theme-**Agriculture, FoodTech & Rural Development
- **PS Category- Hardware**
- **Team ID-**
- **Team Name-AGRI DEVELOPERS**

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KisanSahayak

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❖Proposed Solution (Describe your Idea/Solution/Prototype)

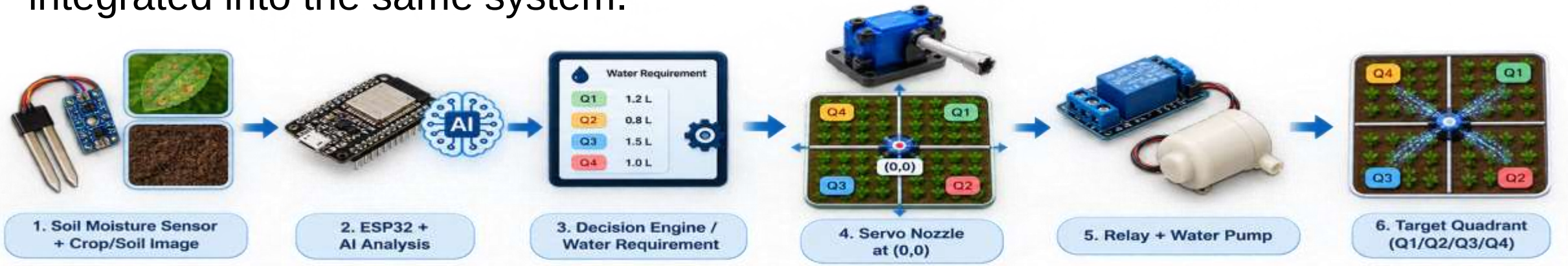
- **AI + ESP32-based** smart farming system for crop health and irrigation monitoring.
- **Soil moisture sensing** measures each farm zone and calculates its individual water requirement.
- **Servo-controlled nozzle at (0,0)** delivers targeted water to the required zone.
- AI-based **crop/soil image analysis** helps identify diseases, pests and nutrient-related issues.
- **Low-cost, modular and scalable** design for precision irrigation and future NPK/environmental sensing.

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TECHNICAL APPROACH

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- **Sensing:** ESP32 + soil-moisture sensor collect zone-wise soil data.
- **Decision Engine:** Moisture readings are processed to calculate the **water requirement of each zone**.
- **Targeting:** Servo motor at **(0,0)** rotates the nozzle toward the required quadrant
- **Irrigation:** Relay-controlled pump delivers the calculated amount of water.
- **AI Analysis:** Mobile/web AI analyzes crop and soil images for **disease, pest and nutrient-stress detection**.
- **Future Scaling:** NPK, environmental sensors and coordinate-level irrigation can be integrated into the same system.



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FEASIBILITY AND VIABILITY

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- **Low-cost:** Uses affordable components such as ESP32, moisture sensor, servo and mini pump.
- **Technically feasible:** Prototype can be built using readily available hardware and simple control logic.
- **Scalable:** Can expand from 4 zones to coordinate-level irrigation with additional sensors.
- **Practical:** Delivers water only where required, reducing unnecessary water usage.
- **Modular:** AI crop analysis, NPK and environmental sensors can be added without redesigning the complete system.
- **Field Potential:** Can be adapted for small farms, nurseries and greenhouse applications.

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- **Water Conservation:** Supplies water according to the actual moisture requirement of each zone.
- **Reduced Input Cost:** Minimizes unnecessary irrigation and resource wastage.
- **Improved Crop Health:** AI-assisted analysis helps identify disease, pest and nutrient-related stress.
- **Better Yield Potential:** Timely irrigation and early detection can support healthier crop growth.
- **Affordable & Accessible:** Low-cost hardware makes the system suitable for small and marginal farmers.
- **Scalable:** Can expand from 4-zone irrigation to larger fields with coordinate-based targeting and additional sensors.

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RESEARCH AND REFERENCES

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Smart India Hackathon 2026 – SIH26180 Problem Statement
SIH 2026 Problem Statements
FAO – Irrigation & Agricultural Water Management
FAO: Irrigation Management
FAO – Water-Use Efficiency in Small-Scale Agriculture
FAO: Field Guide to Improve Water Use Efficiency
Espressif – ESP32 Technical Documentation
ESP32 Technical Documentation
Research – CNNs for Automatic Plant Disease Identification
Plant Disease Detection Research (PMC)